

SAURAV DINESH UPADHYAY

Software Engineer | Java Backend | Spring Boot | Distributed Systems

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Professional Summary

Software Engineer with **4+ years of experience** designing and building scalable backend systems for large-scale AdTech platforms. Core engineer behind a **Real-Time Bidding (RTB) Demand Side Platform (DSP)** serving **~1 billion ad requests/day**, supporting **300+ advertisers** and **8,000+ campaigns**.
Specialized in Java, Spring Boot, distributed systems, Elasticsearch, ClickHouseDB, Aerospike, and production-grade AI integrations. Experienced in designing scalable REST APIs, database schemas, asynchronous processing, search infrastructure, and performance optimization for high-throughput systems.

Technical Skills

- **Languages:** Java, Python, SQL, JavaScript, TypeScript.
- **Backend:** Spring Boot, Spring Security, Spring Data JPA, JWT Authentication, REST APIs, Node.js, FastAPI, Async Processing.
- **Databases:** MySQL, MongoDB, ClickHouseDB, PostgreSQL, Aerospike, Redis.
- **Distributed Systems:** Real-Time Bidding (RTB), Elasticsearch, Logstash, Kafka, Distributed Data Processing, REST Architecture.
- **AI & LLM:** OpenAI API, Ollama, Spring AI, Function Calling, RAG, Prompt Engineering, Structured Outputs, JSON Schema Validation, Text-to-SQL.
- **Tools:** Git, GitLab, Docker, Postman, Swagger, IntelliJ IDEA.

Professional Experience

Mobavenue Pvt Ltd. --- *Software Developer L2*

September 2022 – Present

Backend & Distributed Systems:

- Core engineer in a **4-member team** responsible for building a Real-Time Bidding (RTB) Demand Side Platform serving **~1 billion ad requests/day** across **300+ advertisers** and **8,000+ campaigns**.
- Designed database schemas and implemented **50+ Spring Boot REST APIs** powering campaign management, audience targeting, reporting, analytics, and platform administration.
- Designed secure authentication and authorization using **Spring Security with JWT**.
- Built synchronous and asynchronous backend workflows using Spring's **@Async**.
- Optimized existing APIs and database operations to improve platform responsiveness and maintainability.
- Integrated backend services with React-based applications while maintaining scalable API contracts.

Search Infrastructure:

- Designed and implemented an **Elasticsearch-based product retrieval engine** indexing **10M+ products**.
- Designed index mappings and Logstash ingestion pipelines.
- Achieved average search latency of **25–40 ms** for personalized product retrieval.

User Agent Parser:

- Designed and developed an in-house **User Agent Parsing Engine** replacing a third-party commercial solution.
- Built a hybrid **Tokenization + Regex** parsing engine capable of identifying manufacturer, device model, operating system, and OS version.
- Increased parsing accuracy from **60–70%** to **85–90%** while maintaining comparable execution performance.
- Processes **millions of user-agent strings daily** and eliminates recurring third-party licensing costs.

Audience Manager:

- Optimized ingestion of **70–80 million audience IDs** into Aerospike for real-time bidding workloads.
- Eliminated read-before-write bottlenecks using **Aerospike CDT operations**.
- Enhanced scalability of audience synchronization pipelines under heavy workloads.

AI Engineering:

- Developed AI-powered campaign generation and banner creative generation workflows using **Python and OpenAI APIs**.
- Implemented **Function Calling, Structured Outputs, JSON Schema Validation, Retry Logic, and RAG** for reliable LLM-driven campaign creation.
- Built an AI-powered **Text-to-SQL reporting engine** using Ollama (Qwen), implementing strict validation and response.

Engineering Leadership:

- Owned complete backend delivery from requirement analysis through production deployment.
- Reviewed **50+ pull requests** ensuring engineering quality and consistency.
- Conducted technical interviews for Software Developer hiring.
- Investigated and resolved production issues across backend services.
- Collaborated directly with operations manager for requirement analysis, effort estimation, and feature planning.

Automation:

- Automated reporting and operational workflows using Python and Google Apps Script.
- Reduced manual effort for Operations teams by approximately **3–4 hours per day**.

Projects

Jazor – Production E-commerce Platform

Tech Stack: Node.js, React, Prisma, PostgreSQL, Vercel, Render

- Built and deployed a production-ready perfume e-commerce platform.
- Designed scalable backend APIs and PostgreSQL database schema using Prisma ORM.
- Implemented authentication, product catalog, order management, and responsive user interfaces.

New Sense Store – Microservices Architecture

Tech Stack: Spring Boot, Spring Cloud Gateway, Kafka, Spring Security, Service Discovery

- Developed a microservices-based e-commerce platform to explore distributed system architecture.
- Implemented API Gateway, service discovery, authentication, and asynchronous communication using Kafka.

Sportsphere

Tech Stack: Node.js, PostgreSQL, React Native.

- Designed backend architecture for a real-time sports scoring platform.
- Built REST APIs and optimized relational database schema for match and player statistics.

Education

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| Bachelor of Engineering (B.E.) | Mumbai University 8.3 CGPA 2018-2021 |
| Diploma | Mumbai University 83% 2015 – 2018 |
| SSC (10 th Grade) | Maharashtra Board 81% 2004 – 2015 |